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The bivariate exponentiated additive Weibull distribution and its multivariate extension with applications

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ABSTRACT

Recently, Abd EL-Baset and Ghazal (2020) introduced the exponentiated additive Weibull distribution, which is very useful in modeling the reliability analysis of a product, such as reliability of decision-making and cost–benefit analysis. The products inherently decay with usage and age, resulting in more extreme failures. Fixing a failed product may take longer warranty servicing time, indicating that failure and warranty servicing times correlate to bivariate distributions. This paper introduces a bivariate exponentiated additive Weibull distribution such that the marginals have exponentiated additive Weibull distributions. Statistical properties, parameter estimation using the maximum likelihood, and the bootstrap confidence interval are presented. We also generalize the results for the multivariate case. A simulation study has been executed to evaluate the performance of the parameter estimation via mean square error and absolute bias. Finally, we use three datasets failure and warranty servicing times, American football league, and the air quality to compare the proposed bivariate model to other models. The results revealed that the proposed bivariate model delivers a better fit than other models.

1. Introduction

Inspired by the engineering applications, Weibull (1951) proposed a distribution of broad applicability, which has become one of the significant lifetime distribution in reliability. Weibull distribution is one of the most popular models for analyzing the data in reliability, climate, engineering, biology, and many other fields. For example, Heo et al. (2001) did the flood frequency analysis using the Weibull model, and Wang et al. (1997) applied it in the studies of dielectric failure of multilayer ceramic capacitors. Additionally, two commonly used lifetime models, i.e., Exponential and Rayleigh distributions, are sub-models of the Weibull distribution. Considering the weakness of Weibull distribution in analyzing the data with bathtub-shaped, upside down bathtub-shaped, or bimodal hazard rates. Several developments of the Weibull distribution have been suggested in the literature. For instance, exponentiated Weibull distribution was developed by Mudholkar et al. (1995) to allow the modeling of bathtub-shaped hazard rates. The exponentiated Weibull distribution was obtained by adding one shape parameter to the existing Weibull distribution. Which includes exponentiated exponential (Gupta & Kundu, 2001) and the exponentiated Rayleigh (Kundu & Raqab, 2005) distributions as sub-models. The truncated Weibull distribution was introduced by Zhang and Xie (2011). While the distribution extended Weibull was given by Marshall and Olkin (1997), and Sarhan and Zaindin (2009) developed the modified Weibull distribution. Additionally, some distributions such as the generalized modified Weibull of Carrasco et al. (2008), the exponentiated modified Weibull extension of Sarhan and Apaloo (2013), log-normal modified Weibull by Shakhatreh et al. (2019), a new three-component additive Weibull distribution by Thach (2022), and distribution of modified exponential-Weibull given by Al-Essa et al. (2023) are also studied in the literature.

The additive Weibull (AW) distribution was introduced by Xie and Lai (1996) with cumulative distribution function (CDF)

$$G(u;a,b,c,d) = 1 - e^{-au^b - cu^d}; u \geq 0. \tag{1.1}$$

Assuming that $a, c \geq 0, b > 1, d < 1$, denote it by $U \sim AW(a,b,c,d)$. From the construction, it is clear that the failure rate function has a bathtub shape which involves two Weibull hazards, one is increasing $(b > 1)$ while the other is decreasing $(0 < d < 1)$. Further, the alpha power Weibull distribution is given by Nassar et al. (2017) for modeling the coal-mining disasters data, and along the same line (Abd EL-Baset & Ghazal, 2020) introduced the exponentiated additive Weibull (EAW) distribution for flood peak data with CDF

$$G(x;a,b,c,d,\theta) = \left(1 - e^{-ax^b - cx^d}\right)^{\theta}; \ x \geq 0, \tag{1.2}$$

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where $a, \theta > 0, c \geq 0$ and $b, d > 0$, denote it by $X \sim EAW(a, b, c, d, \theta)$. The EAW distribution is meaningful since it includes numerous sub-models: the additive Weibull, the exponentiated exponential, the exponentiated Weibull distributions, and others. It provides an excellent parametric fit for analyzing data with bathtub failure rates, which give advantages in modeling reliability analysis, such as firmware reliability, reliability of decision-making, and cost–benefit analysis.

Many industrial processes exhibit bivariate characteristics, and a suitable bivariate distribution is desirable to effectively model the bivariate phenomena. The motivation of this paper originates from an article by Park and Kim (2014). The authors have analyzed field data collected from the four nuclear sites in South Korea, 30 failure data are summarized for relatively recent events or failures. They obtained this data from the Operational Performance Information System for Nuclear Power Plants. The data contain the failure times and warranty servicing times which are dependent on the nuclear power plant. With the failure time and the warranty servicing time, we attempt to determine the best-fit distribution for the data. There have been many bivariate extensions of the Weibull as well as other univariate distributions in the literature. For example, the bivariate gamma distribution was given by Nadarajah (2007) for understanding the drought. Meanwhile, the bivariate inverse Weibull distribution was introduced by Muhammed (2016). It was used to analyze the UEFA football data and found that the bivariate inverse Weibull distribution delivers a better fit than the bivariate exponential. The bivariate exponential distribution is utilized by Kim and Rao (2000) to analyze the expected warranty cost of two-attribute free-replacement warranties. Further, El-Morshedy et al. (2020) suggested the bivariate exponentiated discrete Weibull distribution model and discussed its several mathematical and statistical properties. Ramzan et al. (2021) introduced the bivariate extended generalized inverted Kumaraswamy Weibull distribution for modeling the simulated computer series system data of processor and memory lifetimes. The bivariate generalized half-Logistic distribution introduced by Hassan and Chesneau (2022) employs the technique of Farlie–Gumbel–Morgenstern (FGM) copula to measure the financial affordability for Saudi households using the data from the household income and expenditure survey. Davies et al. (2023) extended the bivariate exponential distribution of Downton (1970) to give new bivariate Weibull distribution and used this distribution to model the bone mineral density and car warranty data. El-Morshedy et al. (2023) introduced bivariate exponentiated inverse flexible Weibull extension based on Marshall–Olkin shock model for analyzing the failure times of a parallel system constituted by two identical motors in days. Haj Ahmad et al. (2023) constructed bivariate modified extended Exponential distribution for modeling the data set of 50 simulated primitive computer series systems, each with a processor and a memory they used FGM copula. Fayomi et al. (2023) used the Kavya-Manoharan transformation family of distributions to introduce the FGM bivariate Kavya–Manoharan–Weibull and Clayton bivariate Kavya–Manoharan–Weibull distributions. But so far in the literature, the bivariate extension of the EAW distribution has not been considered.

In this paper we developed the bivariate Exponentiated additive Weibull (BEAW) distribution. Some of the bivariate models like, bivariate generalized linear exponential distribution (Pathak & Vellaisamy, 2020), bivariate generalized exponential distribution (BGED) (Mirhosseini et al., 2015), bivariate generalized Weibull distribution (Pathak et al., 2021) are just the sub-models of the BEAW distribution. The advantage of such construction is that the marginals have exponentiated additive Weibull distributions. Therefore, this extension leads to a more flexible distribution capable of modeling various types of bivariate data. Different properties of this BEAW have been investigated. The parameters of the BEAW distribution are estimated using the maximum likelihood method, and the bootstrap confidence interval is constructed. Next we then consider the multivariate generalization of the BEAW model along the same line and name it as the multivariate Exponentiated additive Weibull (MEAW) model. As expected, the p-variate MEAW model has total $4p + 1$ unknown parameters. The Gibbs sampling method is proposed to generate the data from the proposed BEAW distribution. The simulation study is executed to check the performance of the maximum likelihood estimator based on absolute bias and mean square error. The suggested model was applied to the failure and warranty servicing times, air quality and American football data sets. It inferred that the presented bivariate distribution model better fits the air quality and American football data sets.

The sequence of the article is given as follows: In Section 2, we construct the new bivariate distribution and study its basic statistical properties such as moments and conditional distributions. While Section 3 contains the parameter estimation of the presented bivariate distribution by the maximum likelihood estimation method along with the bootstrap approach to construct the 95% confidence intervals of the parameters. In Section 4 multivariate generalization of the BEAW model is introduced. Section 5 contains the information about random number generation using Gibbs sampling and simulation study to assess the behavior of parameters obtained using the ML estimation. In Section 6 committed to the data analysis of three real-life datasets. Finally, we conclude the paper in Section 7.

2. Model formulation and basic properties

In this section we will define our model and study some special cases and properties of the model. Let $\{\mathscr{U}_1, \mathscr{U}_2, \ldots\}$ and $\{\mathscr{V}_1, \mathscr{V}_2, \ldots\}$ denotes the two sequences of mutually independent and identically distributed (i.i.d.) random variables, where $\mathscr{U}_k \sim \text{AW}(a_1, b_1, c_1, d_1)$, and $\mathscr{V}_k \sim \text{AW}(a_2, b_2, c_2, d_2)$ for $k \in \mathbb{N}$. So, $P(\mathscr{U}_k > x) = e^{-(a_1 x^{b_1} + c_1 x^{d_1})}$ and $P(\mathscr{V}_k > y) = e^{-(a_2 y^{b_2} + c_2 y^{d_2})}$. Define $X = \min\{\mathscr{U}_1, \ldots, \mathscr{U}_N\}$ and $Y = \min\{\mathscr{V}_1, \ldots, \mathscr{V}_N\}$, where N is the random variable. Let N denote the number of trials on which the first success occurs and in the sequence of independent Bernoulli trials the $k-$th trial has probability of success $\frac{\theta}{k}$, $0 < \theta \leq 1, k \in \mathbb{N}$. Then Sibuya distribution will be followed by N with probability mass function (Sibuya, 1979)

$$P(N = n) = \frac{(-1)^{n-1}}{n!}\theta(\theta - 1)\ldots(\theta - n + 1),$$

for $n = 1, 2, \ldots$, and the probability generating function of N is given by

$$h_N(t) = E(t^N) = 1 - (1 - t)^{\theta};\ t \in [0, 1].$$

Then, the survival functions of X and Y are given by

$$\begin{aligned}\bar{G}(x) = P(X > x) &= \sum_{n=1}^{\infty} \left(P(\mathscr{U}_i > x)\right)^n P(N = n) \\ &= \sum_{n=1}^{\infty} \left(e^{-\left(a_1 x^{b_1} + c_1 x^{d_1}\right)}\right)^n P(N = n).\end{aligned}$$

Using the probability generating function of N we have

$$\bar{G}(x) = 1 - \left(1 - e^{-\left(a_1 x^{b_1} + c_1 x^{d_1}\right)}\right)^{\theta}. \tag{2.1}$$

Similarly,

$$\bar{G}(y) = 1 - \left(1 - e^{-\left(a_2 y^{b_2} + c_2 y^{d_2}\right)}\right)^{\theta}. \tag{2.2}$$

It is clear that $X \sim EAW(a_1, b_1, c_1, d_1, \theta)$ and $Y \sim EAW(a_2, b_2, c_2, d_2, \theta)$. The Eqs. (2.1) and (2.2), represent the survival function of EAW random variable follows the CDF of Eq. (1.2). Now, the joint survival function of (X, Y) is given by

$$\begin{aligned}\bar{G}(x, y) = P(X > x, Y > y) &= \sum_{n=1}^{\infty} [P(\mathscr{U}_i > x)P(\mathscr{V}_i > y)]^n P(N = n) \\ &= \sum_{n=1}^{\infty} \left(e^{-\left(a_1 x^{b_1} + c_1 x^{d_1} + a_2 y^{b_2} + c_2 y^{d_2}\right)}\right)^n P(N = n).\end{aligned}$$

Using the probability generating function of N the joint survival function of (X,Y) is given by

$$\bar{G}(x,y) = 1 - \left(1 - e^{-\left(a_1x^{b_1}+c_1x^{d_1}+a_2y^{b_2}+c_2y^{d_2}\right)}\right)^{\theta}. \tag{2.3}$$

Therefore, the joint CDF of (X,Y) is given by

$$G(x,y) = \left(1 - e^{-\left(a_1x^{b_1}+c_1x^{d_1}\right)}\right)^{\theta} + \left(1 - e^{-\left(a_2y^{b_2}+c_2y^{d_2}\right)}\right)^{\theta} - \left(1 - e^{-\left(a_1x^{b_1}+c_1x^{d_1}+a_2y^{b_2}+c_2y^{d_2}\right)}\right)^{\theta};\ \forall\ x,y \in [0,\infty), \tag{2.4}$$

where $a_i, c_i \geq 0, 0 < \theta \leq 1$ and $b_i, d_i > 0; i = 1,2$. The random vector (X,Y) whose joint CDF is given by Eq. (2.4) is said to be bivariate Exponentiated additive Weibull (BEAW) distribution with parameter vector $\bigtriangledown = (a_1, b_1, c_1, d_1, a_2, b_2, c_2, d_2, \theta)$ and will be denoted by $(X,Y) \sim$ BEAW($\bigtriangledown$). The proposed BEAW distribution model has many known distribution functions as special cases. For example,

1. When $b_1 = b_2 = 2$, $c_1 = c_2 = 0$, $a_1 = \frac{\alpha_1}{2}$ and $a_2 = \frac{\alpha_2}{2}$. Then, (2.4) reduced to

   $$G(x,y) = \left(1 - e^{-\frac{\alpha_1}{2}x^2}\right)^{\theta} + \left(1 - e^{-\frac{\alpha_2}{2}y^2}\right)^{\theta} - \left(1 - e^{-\left(\frac{\alpha_1}{2}x^2+\frac{\alpha_2}{2}y^2\right)}\right)^{\theta}.$$

   The above mentioned equation is the joint CDF of bivariate generalized Rayleigh distribution (BGRD). This bivariate distribution model is also discussed in Pathak and Vellaisamy (2020) and $x,y \geq 0, \alpha_1, \alpha_2 > 0, 0 < \theta \leq 1$.
2. For $b_1 = b_2 = 1$, $d_1 = d_2 = 2$, $c_1 = \frac{\beta_1}{2}$ and $c_2 = \frac{\beta_2}{2}$. Then, from (2.4) we get

   $$G(x,y) = \left(1 - e^{-\left(a_1x+\frac{\beta_1}{2}x^2\right)}\right)^{\theta} + \left(1 - e^{-\left(a_2y+\frac{\beta_2}{2}y^2\right)}\right)^{\theta} - \left(1 - e^{-\left(a_1x+\frac{\beta_1}{2}x^2+a_2y+\frac{\beta_2}{2}y^2\right)}\right)^{\theta},$$

   where $x,y \geq 0, a_1, a_2, \beta_1, \beta_2 \geq 0, a_i + \beta_i > 0; i = 1,2$ and $0 < \theta \leq 1$. We refer it to as bivariate generalized linear exponential distribution (Pathak & Vellaisamy, 2020).
3. Again, put $c_1 = c_2 = 0$ and $b_1 = b_2 = 1$, (2.4) corresponds to

   $$G(x,y) = (1 - e^{-a_1x})^{\theta} + (1 - e^{-a_2y})^{\theta} - \left(1 - e^{-(a_1x+a_2y)}\right)^{\theta}.$$

   The above distribution function represents the CDF of bivariate generalized exponential distribution (BGED) (Mirhosseini et al., 2015) with $x,y \geq 0, a_1, a_2 > 0, 0 < \theta \leq 1$.
4. Put $c_1 = c_2 = 0$ and $b_1 = b_2 = b$ in Eq. (2.4), we get bivariate generalized Weibull distribution proposed by Pathak et al. (2021). The CDF of this model is given by

   $$G(x,y) = \left(1 - e^{-a_1x^b}\right)^{\theta} + \left(1 - e^{-a_2y^b}\right)^{\theta} - \left(1 - e^{-(a_1x^b+a_2y^b)}\right)^{\theta},$$

   where $x,y \geq 0, a_1, a_2, b > 0$ and $0 < \theta \leq 1$.

Further, the joint probability density function (PDF) of random vector (X,Y) follows the (2.4) is given by

$$g(x,y) = \theta\left(a_1b_1x^{b_1-1} + c_1d_1x^{d_1-1}\right)\left(a_2b_2y^{b_2-1} + c_2d_2y^{d_2-1}\right) \times e^{-\left(a_1x^{b_1}+c_1x^{d_1}+a_2y^{b_2}+c_2y^{d_2}\right)}\left(1 - \theta e^{-\left(a_1x^{b_1}+c_1x^{d_1}+a_2y^{b_2}+c_2y^{d_2}\right)}\right) \times \left(1 - e^{-\left(a_1x^{b_1}+c_1x^{d_1}+a_2y^{b_2}+c_2y^{d_2}\right)}\right)^{\theta-2}. \tag{2.5}$$

Figs. 1–3 show the CDF and PDF of the BEAW($\bigtriangledown$) distribution for different combinations of parameters. Let us define $\delta(x) = \left(a_1x^{b_1} + c_1x^{d_1}\right)$, $\delta'(x) = \left(a_1b_1x^{b_1-1} + c_1d_1x^{d_1-1}\right)$, $\delta(y) = \left(a_2y^{b_2} + c_2y^{d_2}\right)$, $\delta'(y) = \left(a_2b_2y^{b_2-1} + c_2d_2y^{d_2-1}\right)$, and $\phi(x,y) = \left(a_1x^{b_1} + c_1x^{d_1} + a_2y^{b_2} + c_2y^{d_2}\right)$. From now on we will use these notation throughout this paper. We can write joint survival function in Eq. (2.3) as below

$$\bar{G}(x,y) = 1 - \left(1 - e^{-\phi(x,y)}\right)^{\theta} = \sum_{j=1}^{\infty}\binom{\theta}{j}(-1)^{j+1}e^{-j\phi(x,y)} \tag{2.6}$$

Now, with the help of Eq. (2.6) we get the joint PDF as

$$g(x,y) = \delta'(x)\delta'(y)\sum_{j=1}^{\infty}\binom{\theta}{j}(-1)^{j+1}j^2e^{-j\phi(x,y)} \tag{2.7}$$

2.1. Statistical properties

In industrial engineering, process optimization, quality control, methods for data analysis and decision-making, among others, are very important. They provide engineers and professionals the ability to detect patterns, identify anomalies, evaluate performance, and implement efficient strategies for boosting effectiveness, productivity, and overall operational excellence. In this section, we study some of the properties of the BEAW distribution which make the proposed bivariate distribution more flexible and convenient to use in real-life applications. We drive expression for marginal and conditional distributions. The marginal distribution is often employed for investigating and understanding how specific element of a system or process respond. Meanwhile, conditional distribution can be used to analyze the lifespan of components or systems given certain operating settings or environmental factors.

Theorem 2.1.

1. *Let $(X,Y) \sim BEAW(\bigtriangledown)$. Then marginals distribution of $X \sim EAW(a_1, b_1, c_1, d_1, \theta)$ and $Y \sim EAW(a_2, b_2, c_2, d_2, \theta)$.*
2. *Let $(X,Y) \sim BEAW(\bigtriangledown)$. If $\theta = 1$ then X and Y are independent random variables with $X \sim AW(a_1, b_1, c_1, d_1)$ and $Y \sim AW(a_2, b_2, c_2, d_2)$.*

Proof. (1) From (2.4) the marginal CDFs of X and Y are $G(x) = \lim_{y\to\infty} G(x,y) = \left(1 - e^{-\delta(x)}\right)^{\theta}$ and $G(y) = \lim_{x\to\infty} G(x,y) = \left(1 - e^{-\delta(y)}\right)^{\theta}$, respectively. This implies that marginal CDFs of X and Y follow the EAW distribution in (1.2).

(2) It is trivial. The details are avoided. ■

Theorem 2.2. *Let $(X,Y) \sim BEAW(\bigtriangledown)$. Then,*

1. *Given that $Y = y$, the conditional density of X is represented as*

   $$g(x|y) = \delta'(x)e^{-\delta(x)}\left(1 - \theta e^{-\phi(x,y)}\right)\left(1 - e^{-\phi(x,y)}\right)^{\theta-2}\left(1 - e^{-\delta(y)}\right)^{1-\theta}. \tag{2.8}$$

2. *Given that $X = x$, the conditional density of Y is represented as*

   $$g(y|x) = \delta'(y)e^{-\delta(y)}\left(1 - \theta e^{-\phi(x,y)}\right)\left(1 - e^{-\phi(x,y)}\right)^{\theta-2}\left(1 - e^{-\delta(x)}\right)^{1-\theta}. \tag{2.9}$$

Proof. It is trivial. The details are avoided. ■

Statistical moments provide valuable insights into the characteristics of data distributions, allowing for comparisons, and the identification of specific features. We derive a joint expectation of (X,Y) for BEAW distribution. Industrial engineers use statistical methods to monitor and improve the quality of manufacturing processes. The joint expectation is employed to estimate the mean or expected value of multiple quality-related variables simultaneously, allowing for a comprehensive assessment of process performance.

Theorem 2.3. *Let $(X,Y) \sim BEAW(\bigtriangledown)$. Then*

$$E(x^iy^j) = \sum_{j=1}^{\infty}\binom{\theta}{j}(-1)^{j+1}j^2\sum_{p=0}^{\infty}\frac{(-jc_1)^p}{p!}\left[\frac{a_1\Gamma\left(\frac{l+pd_1+b_1}{b_1}\right)}{(ja_1)^{(l+pd_1+b_1)/b_1}} + \frac{c_1d_1\Gamma\left(\frac{l+pd_1+d_1}{b_1}\right)}{b_1(ja_1)^{(l+pd_1+d_1)/b_1}}\right] \times \sum_{q=0}^{\infty}\frac{(-jc_2)^q}{q!}\left[\frac{a_2\Gamma\left(\frac{l+qd_2+b_2}{b_2}\right)}{(ja_2)^{(l+qd_2+b_2)/b_2}} + \frac{c_2d_2\Gamma\left(\frac{l+qd_2+d_2}{b_2}\right)}{b_2(ja_2)^{(l+qd_2+d_2)/b_2}}\right].$$

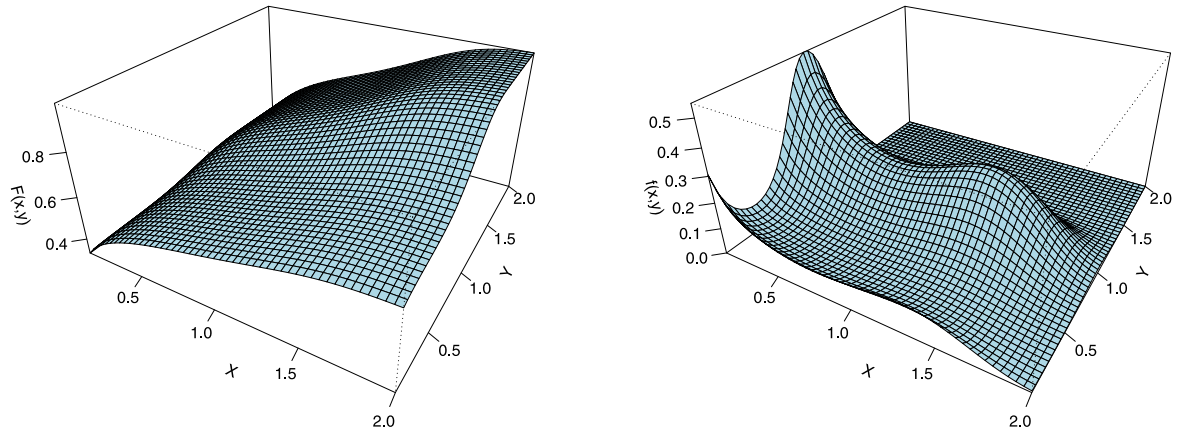


Fig. 1. Probability distribution (left) and density (right) plots for $a_1 = 0.1, a_2 = 0.9, b_1 = 5.5, b_2 = 5.5, c_1 = 0.8, c_2 = 0.8, d_1 = 0.7, d_2 = 0.1, \theta = 0.5$.

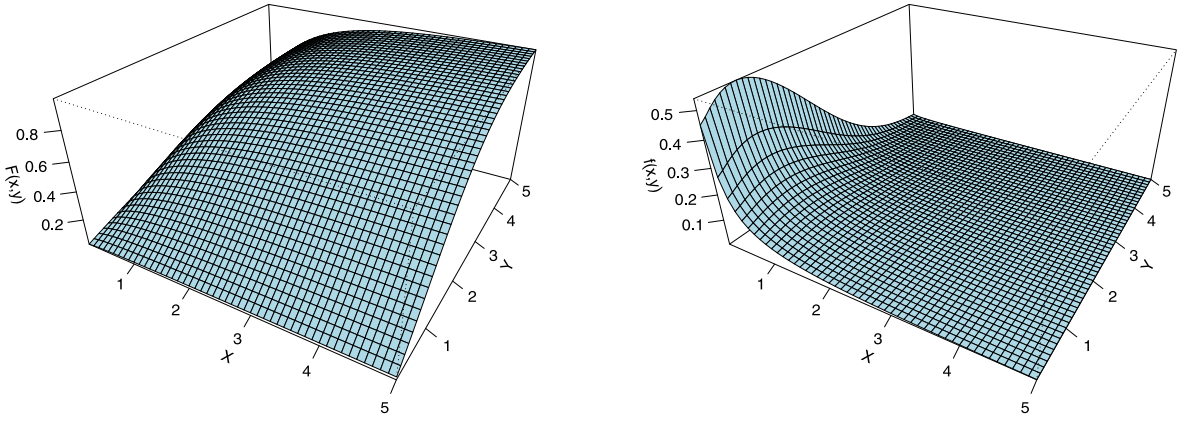


Fig. 2. Probability distribution (left) and density (right) plots for $a_1 = 0.5, a_2 = 0.2, b_1 = 1.1, b_2 = 1.9, c_1 = 0.9, c_2 = 0.2, d_1 = 0.5, d_2 = 0.9, \theta = 0.9$.

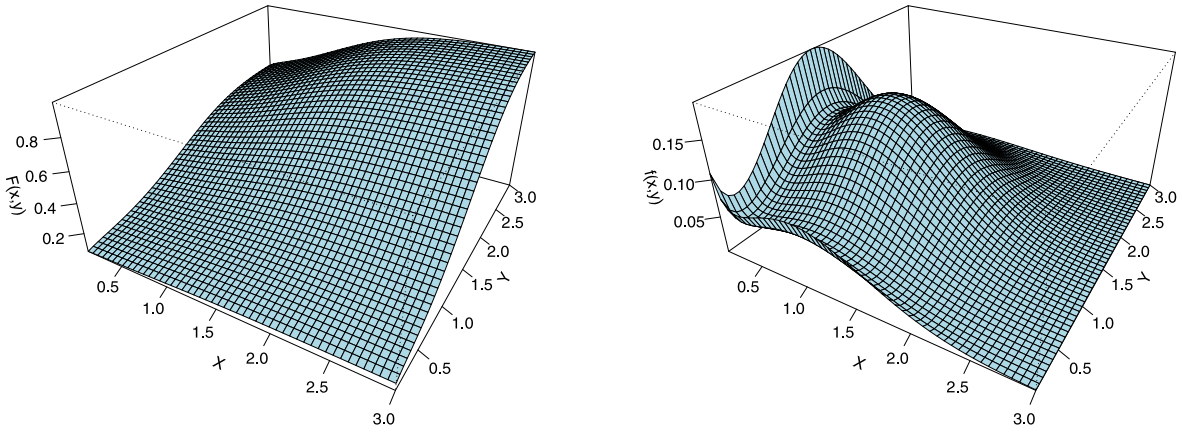


Fig. 3. Probability distribution (left) and density (right) plots for $a_1 = 0.5, a_2 = 0.2, b_1 = 2.3, b_2 = 2.9, c_1 = 0.9, c_2 = 0.2, d_1 = 0.1, d_2 = 0.3, \theta = 0.9$.

Proof. See Appendix. ■

Components in an industrial system need time to time monitoring and maintenance, to evaluate and enhance the reliability and maintainability of systems and processes, industrial engineers frequently utilize the failure rate function (Zhang et al., 2023), which is also referred to as the hazard rate function. The failure rate function offers important information about a system's or component's reliability characteristics. It helps in determining important components that have higher failure rates and assessing the possibility of failure over time (Miller & Childers, 2012). This knowledge may be used by reliability engineers to enhance system reliability overall, optimize maintenance schedules, and enhance system design. The bivariate failure rate function of the BEAW distribution is given below.

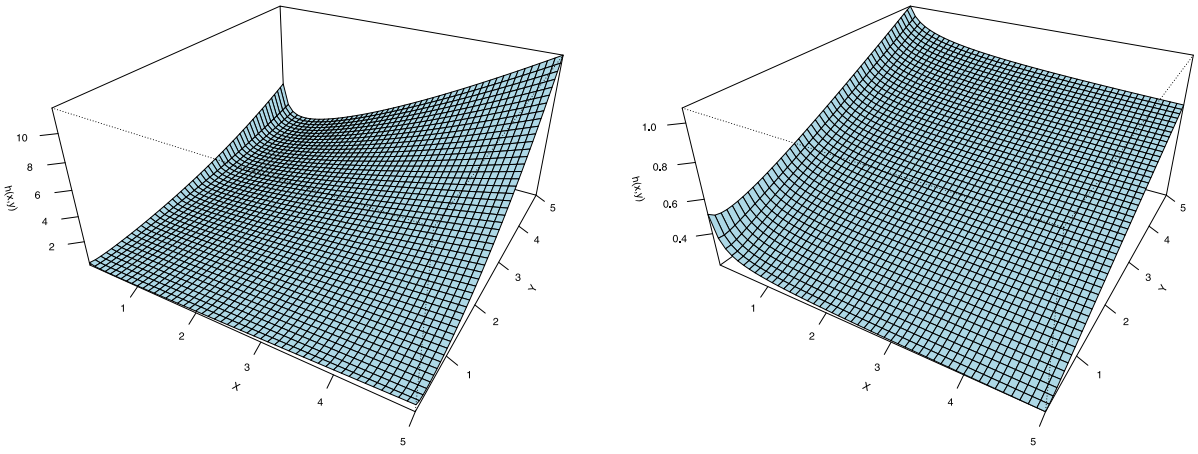


Fig. 4. Plots of hazard rate functions. Left: $a_1 = 0.1, a_2 = 0.1, b_1 = 2.3, b_2 = 2.9, c_1 = 0.9, c_2 = 0.2, d_1 = 0.1, d_2 = 0.3, \theta = 0.9$. Right: $a_1 = 0.1, a_2 = 0.1, b_1 = 1.1, b_2 = 1.9, c_1 = 0.9, c_2 = 0.2, d_1 = 0.9, d_2 = 0.9, \theta = 0.9$.

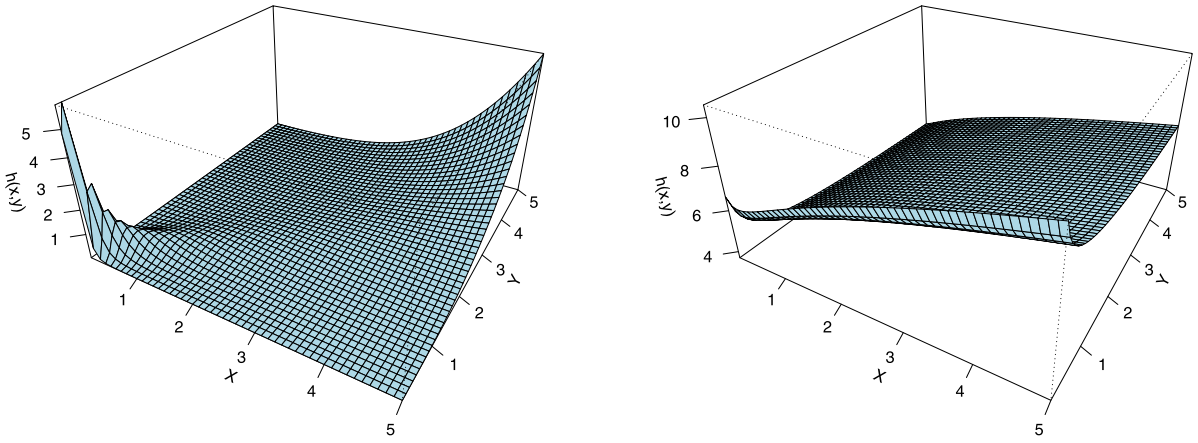


Fig. 5. Plots of hazard rate functions. Left: $a_1 = 0.0001, a_2 = 0.0005, b_1 = 5.5, b_2 = 5.5, c_1 = 0.003, c_2 = 0.004, d_1 = 3, d_2 = 4, \theta = 0.09$. Right: $a_1 = 1.1, a_2 = 0.9, b_1 = 1.5, b_2 = 0.9, c_1 = 1.9, c_2 = 0.9, d_1 = 0.9, d_2 = 0.9, \theta = 0.9$.

Theorem 2.4. *Let $(X, Y) \sim BEAW(\bigtriangledown)$. Then, the bivariate failure rate function, i.e. $h(x, y)$, is defined as*

$$h(x,y) = \frac{\theta\delta'(x)\delta'(y)e^{-\phi(x,y)}\left(1-\theta e^{-\phi(x,y)}\right)\left(1-e^{-\phi(x,y)}\right)^{\theta-2}}{\left(1-\left(1-e^{-\phi(x,y)}\right)^{\theta}\right)}, \tag{2.10}$$

Proof. For a random vector (X, Y) with the survival function in Eq. (2.3) and joint PDF in Eq. (2.5) we will get the bivariate failure rate function $h(x, y)$ which completes the proof. ■

In particular, for $\theta = 1$, Eq. (2.10) reduce to $h(x, y) = \delta'(x)\delta'(y)$ which is the product of bivariate failure rate functions of two independent random variables having the distribution in Eq. (1.1). Also, the plots of bivariate failure rate functions are given in Figs. 4 and 5 with different parameter combination.

3. Parameter estimation

The present section, is devoted to parameters estimation of the BEAW($\bigtriangledown$) distribution model using maximum likelihood (ML) estimation and construction of the 95% confidence intervals for the parameters using the bootstrap method.

3.1. Maximum likelihood estimation

Let $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ be a random sample from BEAW($\bigtriangledown$) model. Now the log-likelihood ($\mathscr{L}$) function of the BEAW($\bigtriangledown$) model using the (2.5) can be written as

$$\begin{aligned}\mathscr{L} &= n\log(\theta) + \sum_{j=1}^{n}\log\left(\delta'(x_j)\right) + \sum_{j=1}^{n}\log\left(\delta'(y_j)\right)\\ &\quad - \sum_{j=1}^{n}\phi(x_j, y_j) + \sum_{j=1}^{n}\log\left(1-\theta e^{-\phi(x_j,y_j)}\right)\\ &\quad + (\theta-2)\sum_{j=1}^{n}\log\left(1-e^{-\phi(x_j,y_j)}\right).\end{aligned} \tag{3.1}$$

In order to obtain the ML estimates, i.e. $(\hat{a_1}, \hat{b_1}, \hat{c_1}, \hat{d_1}, \hat{a_2}, \hat{b_2}, \hat{c_2}, \hat{d_2}, \hat{\theta})$, of $(a_1, b_1, c_1, d_1, a_2, b_2, c_2, d_2, \theta)$. Differentiate Eq. (3.1) with respect to each parameter and equate to zero to get the normal equations. The normal equations are given in Appendix. Eqs. (A.1)–(A.9) given in the Appendix are solved simultaneously to obtain the ML estimates $\hat{\bigtriangledown} = (\hat{a_1}, \hat{b_1}, \hat{c_1}, \hat{d_1}, \hat{a_2}, \hat{b_2}, \hat{c_2}, \hat{d_2}, \hat{\theta})$. ML estimates are not obtained in explicit form; therefore, their solutions are numerically obtained using some optimization methods. We used the optim() function in the *R* software for the same via a good set of initial values.

3.2. Bootstrap confidence interval

Further, to calculate the asymptotic confidence interval it is required to get the observed Fisher information matrix. In order to obtain the observed Fisher information matrix we need to take $9(9+1)/2 = 45$ times derivatives of normal Equations in (A.1)–(A.9) (see Box I), at

$$
\boldsymbol{I} = \begin{bmatrix}
-\frac{\partial^2 \mathscr{L}}{\partial a_1^2} & -\frac{\partial^2 \mathscr{L}}{\partial a_1 \partial b_1} & -\frac{\partial^2 \mathscr{L}}{\partial a_1 \partial c_1} & -\frac{\partial^2 \mathscr{L}}{\partial a_1 \partial d_1} & -\frac{\partial^2 \mathscr{L}}{\partial a_1 \partial a_2} & -\frac{\partial^2 \mathscr{L}}{\partial a_1 \partial b_2} & -\frac{\partial^2 \mathscr{L}}{\partial a_1 \partial c_2} & -\frac{\partial^2 \mathscr{L}}{\partial a_1 \partial d_2} & -\frac{\partial^2 \mathscr{L}}{\partial a_1 \partial \theta} \\
 & -\frac{\partial^2 \mathscr{L}}{\partial b_1^2} & -\frac{\partial^2 \mathscr{L}}{\partial b_1 \partial c_1} & -\frac{\partial^2 \mathscr{L}}{\partial b_1 \partial d_1} & -\frac{\partial^2 \mathscr{L}}{\partial b_1 \partial a_2} & -\frac{\partial^2 \mathscr{L}}{\partial b_1 \partial b_2} & -\frac{\partial^2 \mathscr{L}}{\partial b_1 \partial c_2} & -\frac{\partial^2 \mathscr{L}}{\partial b_1 \partial d_2} & -\frac{\partial^2 \mathscr{L}}{\partial b_1 \partial \theta} \\
 & & -\frac{\partial^2 \mathscr{L}}{\partial c_1^2} & -\frac{\partial^2 \mathscr{L}}{\partial c_1 \partial d_1} & -\frac{\partial^2 \mathscr{L}}{\partial c_1 \partial a_2} & -\frac{\partial^2 \mathscr{L}}{\partial c_1 \partial b_2} & -\frac{\partial^2 \mathscr{L}}{\partial c_1 \partial c_2} & -\frac{\partial^2 \mathscr{L}}{\partial c_1 \partial d_2} & -\frac{\partial^2 \mathscr{L}}{\partial c_1 \partial \theta} \\
 & & & -\frac{\partial^2 \mathscr{L}}{\partial d_1^2} & -\frac{\partial^2 \mathscr{L}}{\partial d_1 \partial a_2} & -\frac{\partial^2 \mathscr{L}}{\partial d_1 \partial b_2} & -\frac{\partial^2 \mathscr{L}}{\partial d_1 \partial c_2} & -\frac{\partial^2 \mathscr{L}}{\partial d_1 \partial d_2} & -\frac{\partial^2 \mathscr{L}}{\partial d_1 \partial \theta} \\
 & & & & -\frac{\partial^2 \mathscr{L}}{\partial a_2^2} & -\frac{\partial^2 \mathscr{L}}{\partial a_2 \partial b_2} & -\frac{\partial^2 \mathscr{L}}{\partial a_2 \partial c_2} & -\frac{\partial^2 \mathscr{L}}{\partial a_2 \partial d_2} & -\frac{\partial^2 \mathscr{L}}{\partial a_2 \partial \theta} \\
 & & & & & -\frac{\partial^2 \mathscr{L}}{\partial b_2^2} & -\frac{\partial^2 \mathscr{L}}{\partial b_2 \partial c_2} & -\frac{\partial^2 \mathscr{L}}{\partial b_2 \partial d_2} & -\frac{\partial^2 \mathscr{L}}{\partial b_2 \partial \theta} \\
 & & & & & & -\frac{\partial^2 \mathscr{L}}{\partial c_2^2} & -\frac{\partial^2 \mathscr{L}}{\partial c_2 \partial d_2} & -\frac{\partial^2 \mathscr{L}}{\partial c_2 \partial \theta} \\
 & & & & & & & -\frac{\partial^2 \mathscr{L}}{\partial d_2^2} & -\frac{\partial^2 \mathscr{L}}{\partial d_2 \partial \theta} \\
 & & & & & & & & -\frac{\partial^2 \mathscr{L}}{\partial \theta^2}
\end{bmatrix},
$$

Box I.

$$
\Sigma_{9\times 9} = \begin{bmatrix}
V(\hat{a_1}) & Cov(\hat{a_1},\hat{b_1}) & Cov(\hat{a_1},\hat{c_1}) & Cov(\hat{a_1},\hat{d_1}) & Cov(\hat{a_1},\hat{a_2}) & Cov(\hat{a_1},\hat{b_2}) & Cov(\hat{a_1},\hat{c_2}) & Cov(\hat{a_1},\hat{d_2}) & Cov(\hat{a_1},\hat{\theta}) \\
 & V(\hat{b_1}) & Cov(\hat{b_1},\hat{c_1}) & Cov(\hat{b_1},\hat{d_1}) & Cov(\hat{b_1},\hat{a_2}) & Cov(\hat{b_1},\hat{b_2}) & Cov(\hat{b_1},\hat{c_2}) & Cov(\hat{b_1},\hat{d_2}) & Cov(\hat{b_1},\hat{\theta}) \\
 & & V(\hat{c_1}) & Cov(\hat{c_1},\hat{d_1}) & Cov(\hat{c_1},\hat{a_2}) & Cov(\hat{c_1},\hat{b_2}) & Cov(\hat{c_1},\hat{c_2}) & Cov(\hat{c_1},\hat{d_2}) & Cov(\hat{c_1},\hat{\theta}) \\
 & & & V(\hat{d_1}) & Cov(\hat{d_1},\hat{a_2}) & Cov(\hat{d_1},\hat{b_2}) & Cov(\hat{d_1},\hat{c_2}) & Cov(\hat{d_1},\hat{d_2}) & Cov(\hat{d_1},\hat{\theta}) \\
 & & & & V(\hat{a_2}) & Cov(\hat{a_2},\hat{b_2}) & Cov(\hat{a_2},\hat{c_2}) & Cov(\hat{a_2},\hat{d_2}) & Cov(\hat{a_2},\hat{\theta}) \\
 & & & & & V(\hat{b_2}) & Cov(\hat{b_2},\hat{c_2}) & Cov(\hat{b_2},\hat{d_2}) & Cov(\hat{b_2},\hat{\theta}) \\
 & & & & & & V(\hat{c_2}) & Cov(\hat{c_2},\hat{d_2}) & Cov(\hat{c_2},\hat{\theta}) \\
 & & & & & & & V(\hat{d_2}) & Cov(\hat{d_2},\hat{\theta}) \\
 & & & & & & & & V(\hat{\theta})
\end{bmatrix},
$$

Box II.

point $\hat{\bigtriangledown} = (\hat{a_1},\hat{b_1},\hat{c_1},\hat{d_1},\hat{a_2},\hat{b_2},\hat{c_2},\hat{d_2},\hat{\theta})$. Such a large amount of derivatives can increase the chances of calculation mistakes. To avoid this we used the bootstrap method to obtain the 95% confidence intervals of the model parameters for the given data. In the bootstrap method first we obtained the approximate variance–covariance matrix $\Sigma_{9\times 9} = I^{-1}$ of $\hat{\bigtriangledown} = (\hat{a_1},\hat{b_1},\hat{c_1},\hat{d_1},\hat{a_2},\hat{b_2},\hat{c_2},\hat{d_2},\hat{\theta})$ (see Box II): and then the 95% confidence interval is obtained by following steps.

Step 1: Obtain 1000 paired bootstrap samples (x_i, y_i) from the data with replacement.

Step 2: For each bootstrap sample find the ML estimates of $a_1, b_1, c_1, d_1, a_2, b_2, c_2, d_2$, and θ. Now, we have 1000 ML estimates for each parameter.

Step 3: Write the above obtained ML estimates for each parameter in $M_{1000\times 9}$ matrix form. And obtain the $\Sigma_{9\times 9}$ covariance matrix from $M_{1000\times 9}$.

Step 4: The quantity $\hat{\mu_i} \pm 1.96 \times \sqrt{\Sigma[i,i]}$ is required 95% confidence interval for the parameter $\mu_i, i = 1, 2, 3, \ldots, 9$.

4. Multivariate exponentiated additive Weibull distribution

An industrial system consists of multiple components or factors which are responsible for the functioning of the system and to model this a multivariate distribution is useful instead of a bivariate or univariate distribution. Therefore, it is essential to study the multivariate generalization of BEAW distribution with there statistical properties for the reliability analysis. In this section, we will define multivariate Exponentiated Additive Weibull Distribution (MEAW) distribution along the same line as the BEAW distribution.

Definition 4.1. A p-variate random vector $(Y_1, \ldots, Y_p)$ is said to have a MEAW distribution with parameters $\bigtriangledown_p = (a_1, b_1, c_1, d_1, \ldots, a_p, b_p, c_p, d_p, \theta)$, if the joint survival function of $(Y_1, \ldots, Y_p)$ is of the form.

$$
\bar{G}(y_1, \ldots, y_p) = 1 - \left(1 - e^{-\phi(y_1,\ldots,y_p)}\right)^{\theta} = \sum_{j=1}^{\infty} \binom{\theta}{j} (-1)^{j+1} e^{-j\phi(y_1,\ldots,y_p)}, \quad (4.1)
$$

where $\phi(y_1, \ldots, y_p) = \left(a_1 y_1^{b_1} + c_1 y_1^{d_1} + \cdots + a_p y_p^{b_p} + c_p y_p^{d_p}\right)$. Now, with the help of Eq. (4.1) we get the joint PDF as

$$
g(y_1, \ldots, y_p) = \prod_{i=1}^{p} \delta'(y_i) \sum_{j=1}^{\infty} \binom{\theta}{j} (-1)^{j+1} j^p e^{-j\phi(y_1,\ldots,y_p)} \quad (4.2)
$$

where $\delta(y_i) = \left(a_i y_i^{b_i} + c_i y_i^{d_i}\right)$, $\delta'(y_i) = \left(a_i b_i y_i^{b_i - 1} + c_i d_i y_i^{d_i - 1}\right)$. As expected, the p-variate MEAW model has total $4p+1$ unknown parameters.

Theorem 4.1.

1. *Let* $(Y_1, \ldots, Y_p) \sim MEAW(\bigtriangledown_p)$. *Then marginals distribution of* $Y_i \sim EAW(a_i, b_i, c_i, d_i, \theta), \forall\ i = 1, \ldots, p$.
2. *Let* $(Y_1, \ldots, Y_p) \sim MEAW(\bigtriangledown_p)$. *If* $\theta = 1$ *then* $Y_i's$ *are independent random variables with* $Y_i \sim AW(a_i, b_i, c_i, d_i), \forall\ i = 1, \ldots, p$.

Proof. (1) Similarly to Theorem 2.1, the marginal CDF of Y_i are

$$
G(y_i) = \lim_{y_1,\ldots,y_{i-1},y_{i+1},\ldots,y_p \to \infty} G(y_1, \ldots, y_p) = \left(1 - e^{-\delta(y_i)}\right)^{\theta}
$$

Which implies that marginal distribution of $Y_i \sim EAW(a_i, b_i, c_i, d_i, \theta)$, $\forall\ i = 1, \ldots, p$.

(2) It is trivial. The details are avoided. ■

Theorem 4.2. *Let* $(Y_1, \ldots, Y_p) \sim MEAW(\bigtriangledown_p)$. *Then*

Table 1
Results of the simulation study for all three different sample sizes.

Sample size		$\hat{a_1}$	$\hat{b_1}$	$\hat{c_1}$	$\hat{d_1}$	$\hat{a_2}$	$\hat{b_2}$	$\hat{c_2}$	$\hat{d_2}$	$\hat{\theta}$
Case 1										
30	Mean	2.12381	2.17249	2.13113	2.15423	2.08035	2.15057	2.11520	2.11813	0.93379
	MSE	0.13860	0.07432	0.14217	0.08341	0.14396	0.07985	0.11240	0.08572	0.01062
	|Bias|	0.31017	0.22365	0.30990	0.22921	0.30575	0.24043	0.27532	0.24421	0.09168
50	Mean	2.12573	2.14956	2.14980	2.13520	2.19149	2.11659	2.22117	2.10618	0.96658
	MSE	0.09487	0.06103	0.08467	0.06361	0.06913	0.05497	0.06536	0.05578	0.00871
	|Bias|	0.26321	0.20239	0.24495	0.20704	0.21850	0.19083	0.21287	0.19160	0.08962
70	Mean	2.18544	2.15932	2.20956	2.15379	2.21853	2.21292	2.23724	2.19822	0.97344
	MSE	0.05822	0.02897	0.05627	0.03014	0.06704	0.03190	0.06469	0.02958	0.00322
	|Bias|	0.20318	0.13986	0.20049	0.14196	0.21330	0.14675	0.21095	0.13985	0.05082
Case 2										
30	Mean	2.57154	1.67744	2.59943	1.65114	2.71908	1.71819	2.69626	1.68222	0.92957
	MSE	0.34017	0.11460	0.32240	0.12753	0.19001	0.08004	0.28146	0.09737	0.02651
	|Bias|	0.46527	0.28848	0.44349	0.30281	0.33553	0.24222	0.36963	0.25937	0.14945
50	Mean	2.79971	1.73739	2.82088	1.71086	2.75006	1.74895	2.76109	1.72643	0.94486
	MSE	0.16054	0.07685	0.17132	0.08896	0.14972	0.06009	0.16171	0.07537	0.02623
	|Bias|	0.33719	0.23273	0.33837	0.24439	0.30801	0.19849	0.30802	0.21712	0.14970
70	Mean	2.82740	1.78058	2.86291	1.75315	2.75640	1.72076	2.80154	1.71474	0.94457
	MSE	0.13146	0.05778	0.11277	0.05907	0.12636	0.05236	0.10506	0.05427	0.02511
	|Bias|	0.29980	0.20215	0.27795	0.20398	0.29383	0.19717	0.26930	0.20114	0.14706
Case 3										
30	Mean	1.76597	2.29619	1.80480	2.27042	1.76650	2.30764	1.80133	2.28252	0.89559
	MSE	0.08436	0.20263	0.09740	0.23674	0.06932	0.20256	0.06839	0.24111	0.05513
	|Bias|	0.23944	0.38033	0.24591	0.39629	0.21743	0.37603	0.21097	0.40081	0.21408
50	Mean	1.86701	2.36250	1.91996	2.35220	1.79113	2.28121	1.85916	2.27332	0.90210
	MSE	0.05657	0.15437	0.06193	0.16135	0.05406	0.16970	0.05067	0.17169	0.05157
	|Bias|	0.19733	0.31424	0.20615	0.32100	0.18653	0.35603	0.18144	0.35857	0.20604
70	Mean	1.84715	2.37331	1.88678	2.34774	1.82333	2.33023	1.87788	2.29472	0.89275
	MSE	0.05030	0.13751	0.06280	0.14959	0.03958	0.13198	0.04245	0.14860	0.04777
	|Bias|	0.17782	0.30521	0.19659	0.31924	0.16494	0.31540	0.16421	0.33139	0.19782

$$E(y_1^l \dots y_p^l) = \sum_{j=1}^{\infty} \binom{\theta}{j} (-1)^{j+1} j^2 \prod_{i=1}^{p} \sum_{p=0}^{\infty} \frac{(-jc_i)^p}{p!} \left[\frac{a_i \Gamma\left(\frac{l+pd_i+b_i}{b_i}\right)}{(ja_i)^{(l+pd_i+b_i)/b_i}} + \frac{c_i d_i \Gamma\left(\frac{l+pd_i+d_i}{b_i}\right)}{b_i (ja_i)^{(l+pd_i+d_i)/b_i}} \right]$$

Proof. Proof is similar to Theorem 2.3. ■

5. Simulation study

To evaluate the behavior of ML estimates for the BEAW distribution model a Monte Carlo simulation experiment was conducted in this section. The *R* software was used to perform the simulations with 1000 iterations for, i.e. $n = 30(20)70$, three different sample sizes. The mean, mean squared error (MSE), and absolute bias for the ML estimates of each parameter are evaluated for each case. Three different combinations of parameters were taken for simulation as follows: Case 1 : $a_1 = b_1 = c_1 = d_1 = a_2 = b_2 = c_2 = d_2 = 2.2, \theta = 0.9$, Case 2 : $a_1 = c_1 = a_2 = c_2 = 2.9, b_1 = d_1 = b_2 = d_2 = 1.9, \theta = 0.8$, and Case 3 : $a_1 = c_1 = a_2 = c_2 = 1.8, b_1 = d_1 = b_2 = d_2 = 2.6, \theta = 0.7$. Since it is very difficult to draw a random vector from joint CDF in Eq. (2.4). We used the Gibbs sampling technique as follows:

Step 1: Initialize $x_0 = 0.5$ and $y_0 = 0.5$.
Step 2: Generate random variate from conditional density in Eq. (2.8) taking initial value $y_0 = 0.5$ given in step 1.
Step 3: Update the value of x with the new obtained value in step 2.
Step 4: Generate random variate from conditional density in Eq. (2.9) taking the updated value of x_1 given in step 2.
Step 5: Now, the obtained pair (x_1, y_1) is the required random vector.
Step 6: Repeat, steps 2 to 4, $m + n$ number of times. Burn first m variates, we obtain the sample of size n.

Findings are presented in Table 1. We noticed that perhaps the MSE and absolute biases of each of $\hat{a_1}, \hat{b_1}, \hat{c_1}, \hat{d_1}, \hat{a_2}, \hat{b_2}, \hat{c_2}, \hat{d_2}$, and $\hat{\theta}$ decrease as increasing the size of the sample, see Table 1. Also, the mean value of the estimates is very close to the true value of the parameter. This indicates that the ML estimation provides good estimates of the parameters for the BEAW model.

5.1. Simulated data

We generated 60 bivariate observations from the proposed BEAW distribution using the Gibbs sampling technique provided in above section with parameters $a_1 = 0.2, b_1 = 1.2, c_1 = 0.3, d_1 = 1.4, a_2 = 0.2, b_2 = 1.2, c_2 = 0.3, d_2 = 1.4$, and $\theta = 0.8$. Further, the proposed BEAW($\bigtriangledown$) model was fitted to simulated data. For comparison with the BEAW distribution, we used BGED, BGRD, bivariate gamma distribution (BGaD) of Nadarajah (2007), Clayton copula based bivariate Rayleigh distribution (CBRD) discussed by El-Sherpieny and Almetwally (2019), and bivariate Dagum distribution (BVDD) of Popović et al. (2018). The Akaike information criteria (AIC) value calculated as $-2\times\mathscr{L}+2\times K$ was used for goodness of fit test, where K represents the number of parameters in a model. A model with the smallest value of AIC and the largest value of $\mathscr{L}$ was considered better fit. For simulated data, the ML estimates of the parameters, $\mathscr{L}$, and AIC values are given in Table 2. It is observed that for the simulated data proposed BEAW model outperformed the other models which is obvious because the data was generated from the BEAW distribution.

6. Applications

This section is devoted to showing the applicability of the proposed BEAW model in real-life situations. For this purpose three bivariate data sets failure times and warranty servicing times (FWS) for nuclear power plants data (Park & Kim, 2014), American football (AF) data (Jamalizadeh & Kundu, 2013), and Air quality (AQ) data (Chambers et al., 1983) were used in this section. The scatter plots of the considered datasets along with other statistical measures are presented in Fig. 6 and Table 3, respectively. From the scatter plots we observed that the correlation between variables is weak for FWS and AQ data and

Table 2
The ML estimates of the parameters, $\mathscr{L}$, and AIC values for the simulated data.

Model	ML estimates of the parameters	$\mathscr{L}$	AIC
BEAW	$\hat{a_1} = 0.3271, \hat{b_1} = 1.3259, \hat{c_1} = 0.3271, \hat{d_1} = 1.3259$	−146.86	311.72
	$\hat{a_2} = 0.2400, \hat{b_2} = 1.4109, \hat{c_2} = 0.2400, \hat{d_2} = 1.4109, \hat{\theta} = 0.9033$		
BGED	$\hat{a_1} = 0.8343, \hat{a_2} = 0.6910, \hat{\theta} = 1.0000$	−153.04	312.08
BGRD	$\hat{a_1} = 0.5992, \hat{a_2} = 0.4131, \hat{\theta} = 0.6082$	−156.70	319.41
BGaD	$\hat{a} = 1.6686, \hat{b} = 0.4023$	−173.17	350.34
CBRD	$\hat{\lambda}_1 = 0.6323, \hat{\lambda}_2 = 0.5403, \hat{\theta} = 0.1471$	−168.76	343.53
BVDD	$\hat{\beta_1} = 0.3992, \hat{\beta_2} = 0.5917, \hat{\beta_3} = 0.0009$	−152.15	314.31
	$\hat{\lambda} = 2.3919, \hat{\delta} = 2.4506$		

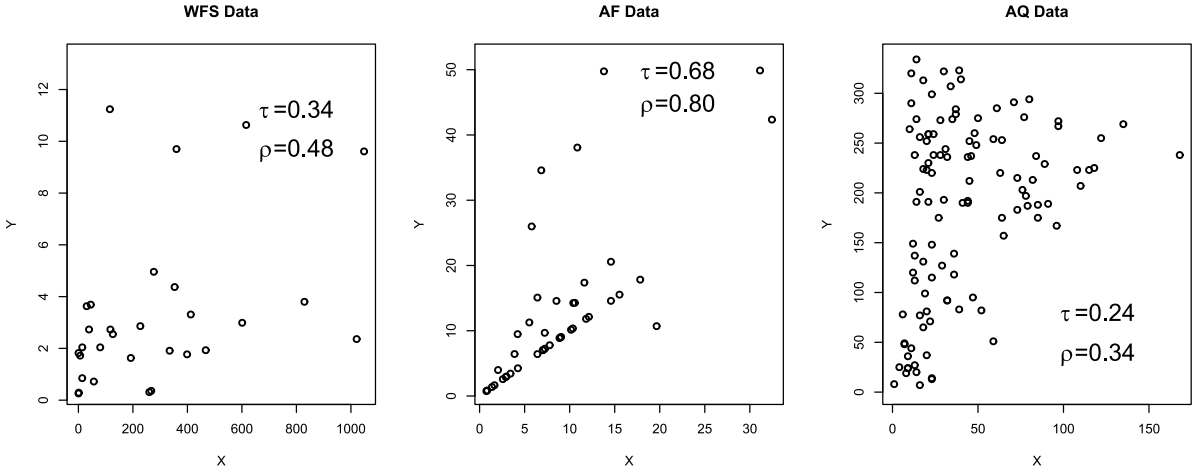


Fig. 6. Scatter plots of WFS, AF, and AQ datasets.

Table 3
Statistical measures for FWS, AF, and AQ datasets.

Measure	FWS		AF		AQ	
	X	Y	X	Y	X	Y
Min	0.35	0.26	0.75	0.75	1.00	7.00
Q_1	40.54	1.73	4.22	6.42	18.00	113.50
Median	209.69	2.45	7.51	9.91	31.00	207.00
Mean	277.05	3.29	9.07	13.42	42.10	184.80
Q_3	389.08	3.67	11.43	14.95	62.00	255.50
Max	1048.23	11.24	32.45	49.88	168.00	334.00
Skewness	1.26	1.54	1.66	1.67	1.24	−0.48
Kurtosis	3.84	4.37	6.36	5.12	4.20	2.07
StDev	296.06	3.04	6.91	12.47	33.27	91.15
Range	1047.88	10.98	31.70	49.13	167.00	327.00

moderate for AF data. From Table 3 it is noticed that all the variables are positively skewed except solar radiation in AQ data which has negative skewness. While the failure time in FWS data has the widest range among all the variables of considered data.

6.1. Failure and warranty servicing times data

In this subsection, the lifetime data on failure times and warranty servicing times for nuclear power plants are considered. For more details about the data see, Park and Kim (2014). Park and Kim (2014) has represented the failure time and warranty servicing times in days. In this data one variable (say X) represents the failure times in days, and another variable (say Y) represents the warranty servicing times in days for nuclear power plants. For convenience, all data points in this data have been divided by 10. The marginal behavior of the failure times and warranty servicing times was examined, EAW distribution was fitted to X and Y separately. After fitting, the log-likelihood $\mathscr{L}$ values, ML estimates, Kolmogorov–Smirnov (KS) test statistic along with p-values are given see Table 4.

Looking at the KS test p-values we could not reject the EAW distribution for the marginals. Now, we fitted the proposed BEAW($\bigtriangledown$) model to failure times and warranty servicing times data. For comparison with the BEAW distribution, we used BGED, BGRD, BGaD, CBRD, and BVDD models. The AIC values and likelihood ratio test were considered as comparison criteria. The quantity $-2\times\mathscr{L}+2\times K$ was used to calculate the AIC value, where K represents the number of parameters in a model. A model with the smallest value of AIC and the largest value of $\mathscr{L}$ was considered better. For failure times and warranty servicing times data, the ML estimates of the parameters, $\mathscr{L}$, and AIC values are given in Table 5. Also, we plot the profiles of the $\mathscr{L}$ of $a_1, b_1, c_1, d_1, a_2, b_2, c_2, d_2, \theta$ in Fig. 7. From Table 5, it is obvious that the BEAW model has the largest $\mathscr{L}$ value and smallest AIC value. Since BGED and BGRD are sub-models of BEAW distribution. We consider the test of hypothesis (i) $H_0 : b_1 = b_2 = 1, c_1 = c_2 = 0$ (BGED) and (ii) $H_0 : b_1 = b_2 = 2, c_1 = c_2 = 0, a_1 = a_1/2, a_2 = a_2/2$ (BGRD) versus H_1 : data follows the BEAW distribution and performs the likelihood ratio test for failure times and warranty servicing times data. For considered models, $2[\mathscr{L}_{BEAW} - \mathscr{L}_{BGED}] = 13.38$ and $2[\mathscr{L}_{BEAW} - \mathscr{L}_{BGRD}] = 76.82$ with corresponding p-values $0.0373 < 0.05$ and approximately zero, respectively, which show that the BEAW distribution provides a significantly better fit over the BGED and BGRD models for the failure times and warranty servicing times data at 5% significance level. Both criteria suggest the BEAW

Table 4
The $\mathscr{L}$ values, ML estimates, and KS statistic with corresponding p-values of X and Y for the failure times and warranty servicing times data.

Marginal	$\mathscr{L}$	$\hat{a}$	$\hat{b}$	$\hat{c}$	$\hat{d}$	$\hat{\theta}$	KS	p-value
X	−124.34	0.2142	0.4081	0.0021	1.5113	1.1651	0.0928	0.93
Y	4.16	2.0987	0.7910	1.8742	0.7911	2.1381	0.1496	0.51

Table 5
The ML estimates of the parameters, $\mathscr{L}$, and AIC values for the failure times and warranty servicing times data.

Model	ML estimates of the parameters	$\mathscr{L}$	AIC
BEAW	$\hat{a_1} = 0.0024, \hat{b_1} = 1.4467, \hat{c_1} = 0.0657, \hat{d_1} = 0.6113$ $\hat{a_2} = 1.6826, \hat{b_2} = 1.4366, \hat{c_2} = 1.4628, \hat{d_2} = 1.3453, \hat{\theta} = 0.7064$	−117.96	253.92
BGED	$\hat{a_1} = 0.0329, \hat{a_2} = 2.4114, \hat{\theta} = 0.7910$	−124.65	255.30
BGRD	$\hat{a_1} = 0.00006, \hat{a_2} = 0.1358, \hat{\theta} = 0.1425$	−157.14	320.29
BGaD	$\hat{a} = 0.0722, \hat{b} = 4.1662$	−204.89	413.79
CBRD	$\hat{\lambda_1} = 0.0243, \hat{\lambda_2} = 2.0090, \hat{\theta} = 0.1785$	−162.07	330.14
BVDD	$\hat{\beta_1} = 37.7674, \hat{\beta_2} = 11.0810, \hat{\beta_3} = 0.00001$ $\hat{\lambda} = 0.0374, \hat{\delta} = 0.5179$	−150.26	310.52

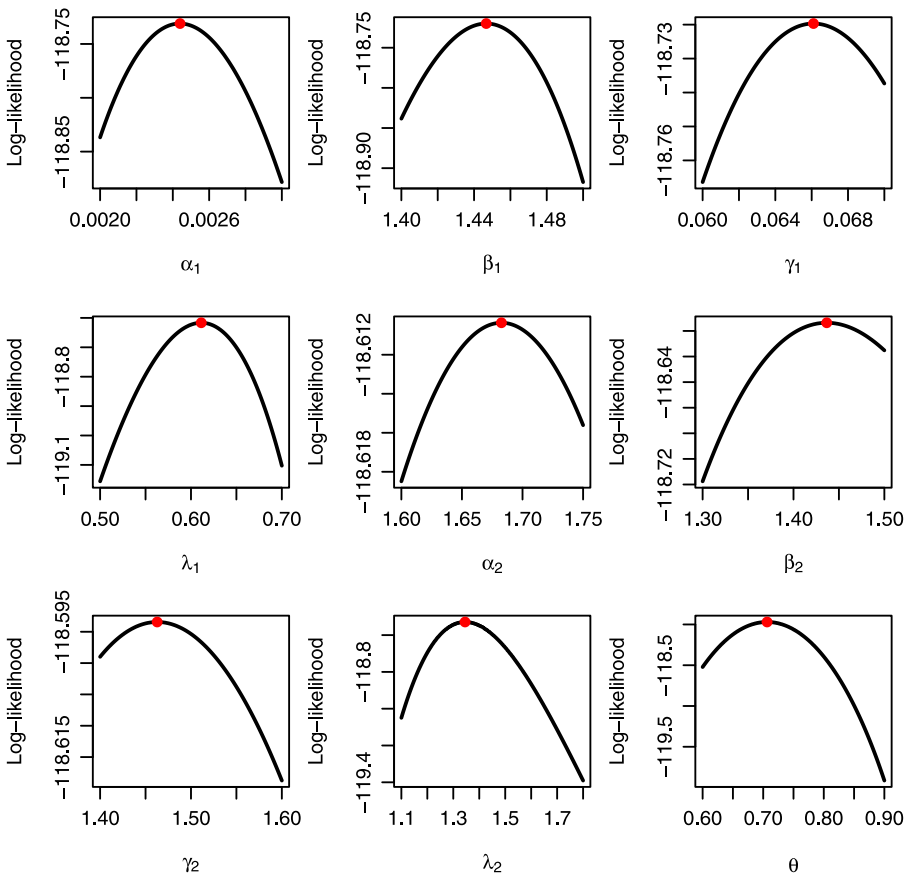


Fig. 7. The curves log-likelihood function of $\bigtriangledown = (a_1, a_2, b_1, b_2, c_1, c_2, d_1, d_2, \theta)$ for failure times and warranty servicing times data.

distribution compared to BGED, BGRD, BGaD, CBRD, and BVDD models for fitting the given failure times and warranty servicing times data.

The approximate 95% confidence intervals for a_1, b_1, c_1, d_1, a_2, b_2, c_2, d_2, and θ are $(-0.0037, 0.0085)$, $(1.2401, 1.6532)$, $(0.0027, 0.1286)$, $(0.3612, 0.8613)$, $(0.4358, 2.9293)$, $(0.9431, 1.9301)$, $(0.2036, 2.7121)$, $(0.6948, 1.9957)$, and $(0.4341, 9786)$, respectively. The covariance matrix, i.e. $\Sigma_{9\times9}^{FWS}$, was used to construct the confidence intervals for failure times and warranty servicing times data (see Box III).

6.2. American football league

The second data is from the American football league, in this set of data, one variable, let us say X, denotes the game's time for the first points earned by kicking the ball between the goal posts, and another, let us say Y, denotes the game's time for moving the ball into the end zone. And this data can be found in Jamalizadeh and Kundu (2013). For computational ease, all data points in American football data have been divided by 10. Again, the marginal behavior of the American football data was examined, and EAW distribution was fitted to X and Y separately. After fitting, the log-likelihood $\mathscr{L}$ values, ML estimates, and KS test statistic along with p-values are given see Table 6. Looking at the KS test p-values we could not reject the EAW distribution for the marginals. Further, the proposed BEAW($\bigtriangledown$) model was fitted to football data, and for comparison with the BEAW distribution, the same models and comparison criteria are considered as in the previous subsection. For football data, the ML estimates of the

$$\Sigma_{9\times9}^{FWS} = \begin{bmatrix} 0.00001 & -0.00020 & -0.00003 & -0.00002 & 0.00037 & 0.00012 & 0.00039 & 0.00014 & 0.00001 \\ & 0.01111 & 0.00126 & -0.00287 & -0.00084 & -0.00792 & -0.00130 & -0.00903 & 0.00138 \\ & & 0.00103 & -0.00057 & 0.00222 & -0.00465 & 0.00223 & -0.00570 & 0.00216 \\ & & & 0.01628 & -0.00466 & 0.00331 & -0.00553 & 0.00085 & 0.00006 \\ & & & & 0.40464 & 0.01659 & 0.40823 & 0.06955 & 0.01031 \\ & & & & & 0.06339 & 0.01570 & 0.07831 & -0.02510 \\ & & & & & & 0.41272 & 0.06938 & 0.01096 \\ & & & & & & & 0.11013 & -0.03154 \\ & & & & & & & & 0.01930 \end{bmatrix}.$$

Box III.

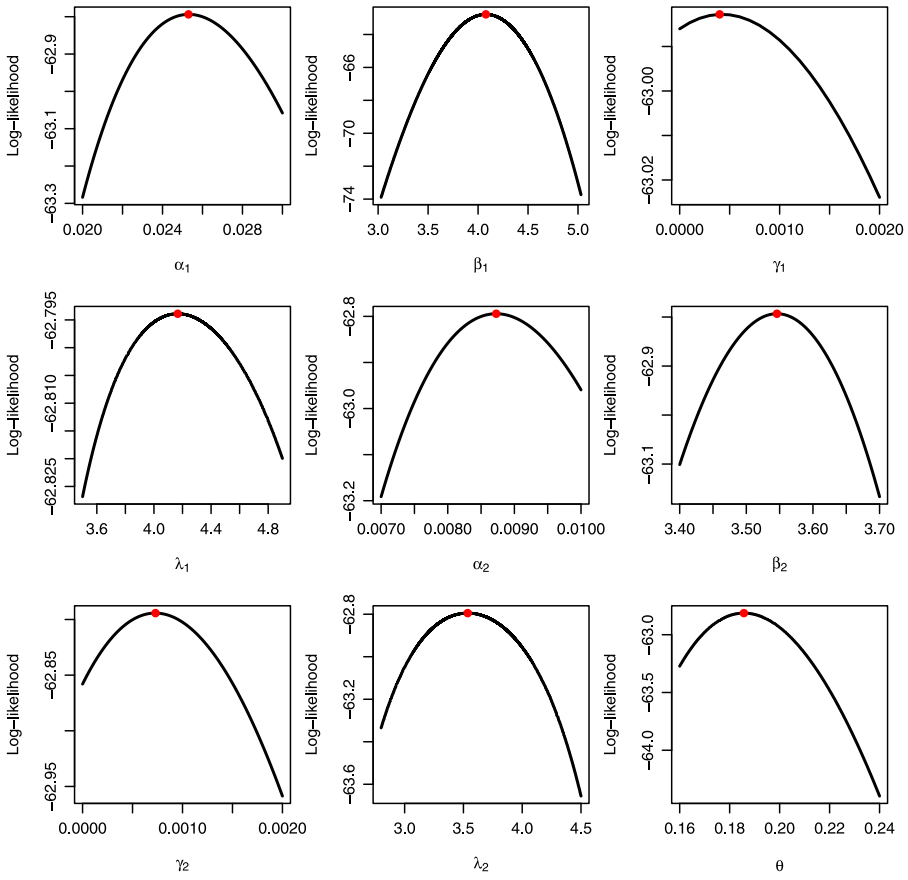


Fig. 8. The curves log-likelihood function of $\bigtriangledown = (a_1, a_2, b_1, b_2, c_1, c_2, d_1, d_2, \theta)$ for American football data.

parameters, $\mathscr{L}$, and AIC values are given in Table 7. Also, we plot the profiles of the $\mathscr{L}$ of $a_1, b_1, c_1, d_1, a_2, b_2, c_2, d_2, \theta$ in Fig. 8. From Table 7, it is obvious that the BEAW model has the largest $\mathscr{L}$ value and smallest AIC value. Both criteria suggest the BEAW distribution compared to BGED, BGRD, BGaD, CBRD, and BVDD models for fitting the given football data. As mentioned the BGED and BGRD are sub-models of BEAW. Again, to conduct the likelihood ratio test for American football data this time, the hypothesis considered are as (i) $H_0 : b_1 = b_2 = 1, c_1 = c_2 = 0$ (BGED) and (ii) $H_0 : b_1 = b_2 = 2, c_1 = c_2 = 0, a_1 = \alpha_1/2, a_2 = \alpha_2/2$ (BGRD) versus H_1 : data follows the BEAW distribution. For considered models, $2[\mathscr{L}_{BEAW} - \mathscr{L}_{BGED}] = 58$ with corresponding p-value approximately zero and $2[\mathscr{L}_{BEAW} - \mathscr{L}_{BGRD}] = 20.14$ with corresponding p-value of 0.00261, which shows that the BEAW distribution model furnishes significantly better fit compared to the BGED and BGRD models for the football data.

The approximate 95% confidence intervals for $a_1, b_1, c_1, d_1, a_2, b_2, c_2, d_2$, and θ are (−0.0837, 0.1349), (3.5019, 4.5754), (−0.0346, 0.0354), (3.3063, 4.7584), (−0.0257, 0.04337), (2.8344, 4.2779), (−0.0053, 0.0069), (2.8298, 4.2675), and (0.1024, 0.2717), respectively. The covariance matrix, i.e. $\Sigma_{9\times9}^{AF}$, was used to construct the confidence intervals for football data (see Box IV).

6.3. Air quality data

The last dataset was collected from May to September 1973 in New York and can be found in Chambers et al. (1983). This data set is named 'airquality' and is openly available in the R package datasets. This data represents the mean ozone in parts per billion (say X) from 1 PM to 3 PM hours at Roosevelt Island and solar radiation (say Y) in Langleys in the frequency band 4000–7700 Angstroms from 8 AM to 12 PM hours at Central Park. In air quality data, we consider 111 observations for the analysis and discarded the missing values. For computational ease, all data points in air quality data have been divided by 100. To study the marginal behavior of the air quality data, EAW distribution was

$$\Sigma_{9\times9}^{AF} = \begin{bmatrix} 0.00311 & -0.00987 & 0.00009 & 0.00179 & 0.00090 & -0.01676 & 0.00001 & -0.01188 & 0.00203 \\ & 0.07500 & -0.00167 & -0.01844 & -0.00377 & 0.08350 & 0.00022 & 0.07526 & -0.00890 \\ & & 0.00032 & 0.00188 & 0.00009 & -0.00259 & 0.00001 & -0.00261 & 0.00030 \\ & & & 0.13721 & 0.00054 & -0.00827 & 0.00021 & -0.00945 & 0.00157 \\ & & & & 0.00031 & -0.00570 & -0.000004 & -0.00437 & 0.00068 \\ & & & & & 0.13561 & 0.00011 & 0.11280 & -0.01440 \\ & & & & & & 0.00001 & 0.00031 & 0.00001 \\ & & & & & & & 0.13451 & -0.01123 \\ & & & & & & & & 0.00186 \end{bmatrix}.$$

Box IV.

Table 6
The $\mathscr{L}$ values, ML estimates, and KS statistic with corresponding p-values of X and Y for the American football data.

Marginal	$\mathscr{L}$	$\hat{a}$	$\hat{b}$	$\hat{c}$	$\hat{d}$	$\hat{\theta}$	KS	p-value
X	−34.12	1.4986	1.0607	0.0127	1.0584	1.7256	0.0941	0.85
Y	−53.03	0.5284	0.7615	0.8639	0.7615	2.3501	0.0991	0.76

Table 7
The ML estimates of the parameters, $\mathscr{L}$, and AIC values for the American football data.

Model	ML estimates of the parameters	$\mathscr{L}$	AIC
BEAW	$\hat{a_1} = 0.0256, \hat{b_1} = 4.0387, \hat{c_1} = 0.0004, \hat{d_1} = 4.0324$ $\hat{a_2} = 0.0088, \hat{b_2} = 3.5562, \hat{c_2} = 0.0008, \hat{d_2} = 3.5487, \hat{\theta} = 0.1871$	−62.79	143.58
BGED	$\hat{a_1} = 1.0331, \hat{a_2} = 0.7082, \hat{\theta} = 0.9159$	−91.99	189.97
BGRD	$\hat{\alpha_1} = 0.3160, \hat{\alpha_2} = 0.1661, \hat{\theta} = 0.4069$	−73.06	152.00
BGaD	$\hat{a} = 2.2038, \hat{b} = 0.6664$	−76.65	157.31
CBRD	$\hat{\lambda_1} = 0.6932, \hat{\lambda_2} = 0.5608, \hat{\theta} = 2.3534$	−71.29	148.58
BVDD	$\hat{\beta_1} = 0.0174, \hat{\beta_2} = 0.2168, \hat{\beta_3} = 0.4237$ $\hat{\lambda} = 2.5677, \hat{\delta} = 2.4951$	−99.85	209.71

Table 8
The log-likelihood $\mathscr{L}$ values, ML estimates, and KS statistic with corresponding p-values of X and Y for the air quality data.

Marginal	$\mathscr{L}$	$\hat{a}$	$\hat{b}$	$\hat{c}$	$\hat{d}$	$\hat{\theta}$	KS	p-value
X	−5.05	0.3782	3.9404	4.2762	0.4841	7.9356	0.0693	0.65
Y	−126.00	0.0057	5.5018	1.0651	0.4028	3.5787	0.0377	0.99

Table 9
The ML estimates of the parameters, $\mathscr{L}$, and AIC values for the air quality data.

Model	ML estimates of the parameters	$\mathscr{L}$	AIC
BEAW	$\hat{a_1} = 1.4883, \hat{b_1} = 1.8485, \hat{c_1} = 0.8178, \hat{d_1} = 1.8552$ $\hat{a_2} = 0.0038, \hat{b_2} = 5.6744, \hat{c_2} = 0.0750, \hat{d_2} = 1.4301, \hat{\theta} = 0.5824$	−123.41	264.82
BGED	$\hat{a_1} = 2.3753, \hat{a_2} = 0.5411, \hat{\theta} = 1.0000$	−194.14	394.27
BGRD	$\hat{\alpha_1} = 2.6891, \hat{\alpha_2} = 0.1671, \hat{\theta} = 0.6422$	−151.25	308.51
BGaD	$\hat{a} = 4.7500, \hat{b} = 0.1515$	−196.51	397.03
CBRD	$\hat{\lambda_1} = 1.8758, \hat{\lambda_2} = 0.4627, \hat{\theta} = 0.4730$	−161.37	328.75
BVDD	$\hat{\beta_1} = 0.0382, \hat{\beta_2} = 0.1231, \hat{\beta_3} = 0.00001$ $\hat{\lambda} = 210430.8, \hat{\delta} = 11.6626$	−205.07	420.15

fitted to X and Y separately. After fitting, the log-likelihood $\mathscr{L}$ values, ML estimates, and KS test statistic along with p-values are given see Table 8. Looking at the KS test p-values we could not reject the EAW distribution for the marginals.

Further, the proposed BEAW($\bigtriangledown$) model was fitted to air quality data. For comparison with the BEAW model, the same BGED, BGRD, BGaD, CBRD, and BVDD models were considered. To select the best model the log-likelihood $\mathscr{L}$ and AIC values were taken into account. For air quality data, the ML estimates of the parameters, $\mathscr{L}$, and AIC values are given in Table 9. To illustrate the likelihood equations, we plot the profiles of the $\mathscr{L}$ of $a_1, b_1, c_1, d_1, a_2, b_2, c_2, d_2, \theta$ in Fig. 9. From Table 9, it is obvious that the BEAW model has the largest $\mathscr{L}$ value and smallest AIC value for air quality data. So, the BEAW distribution is better compared to BGED, BGRD, BGaD, CBRD, and BVDD for fitting the given data. To conduct the likelihood ratio test for air quality data, the hypothesis considered are as (i) $H_0 : b_1 = b_2 = 1, c_1 = c_2 = 0$ (BGED) and (ii) $H_0 : b_1 = b_2 = 2, c_1 = c_2 = 0, a_1 = \alpha_1/2, a_2 = \alpha_2/2$ (BGRD) versus H_1 : data follows the BEAW distribution. For considered models, $2[\mathscr{L}_{BEAW} - \mathscr{L}_{BGED}] = 140.44$ and $2[\mathscr{L}_{BEAW} - \mathscr{L}_{BGRD}] = 54.66$ with corresponding p-value approximately zero. Again, which shows that the BEAW distribution outperformed the BGED and BGRD models for the air quality data. Now we calculate the covariance matrix $\Sigma_{9\times9}^{AQ}$ (see Box V) using the bootstrap procedure given in sub- Section 3.2. Based on $\Sigma_{9\times9}^{AQ}$, the approximate 95% confidence intervals for $a_1, b_1, c_1, d_1, a_2, b_2, c_2, d_2$, and θ are (0.8508, 2.1257), (1.4101, 2.2868), (0.0823, 1.5532), (1.1081, 2.6022), (−0.0211, 0.0284), (4.6578, 6.6908), (0.0188, 0.1311), (0.8335, 2.0266), and (0.4136, 0.7511), respectively.

$$\Sigma^{AQ}_{9\times9} = \begin{bmatrix} 0.10578 & 0.00692 & 0.08035 & 0.03414 & -0.00035 & 0.01694 & 0.00264 & -0.01223 & 0.01225 \\ & 0.05002 & 0.00878 & 0.03085 & 0.00036 & -0.02671 & -0.00374 & 0.02565 & -0.01171 \\ & & 0.14080 & -0.03511 & 0.00004 & -0.01582 & 0.00361 & -0.00532 & 0.00876 \\ & & & 0.14527 & -0.00029 & 0.02290 & -0.00406 & 0.02754 & -0.01246 \\ & & & & 0.00016 & -0.00543 & -0.00012 & -0.00093 & -0.00010 \\ & & & & & 0.26902 & 0.00522 & 0.00910 & 0.00781 \\ & & & & & & 0.00082 & -0.00202 & 0.00167 \\ & & & & & & & 0.09264 & -0.01253 \\ & & & & & & & & 0.00741 \end{bmatrix}.$$

Box V.

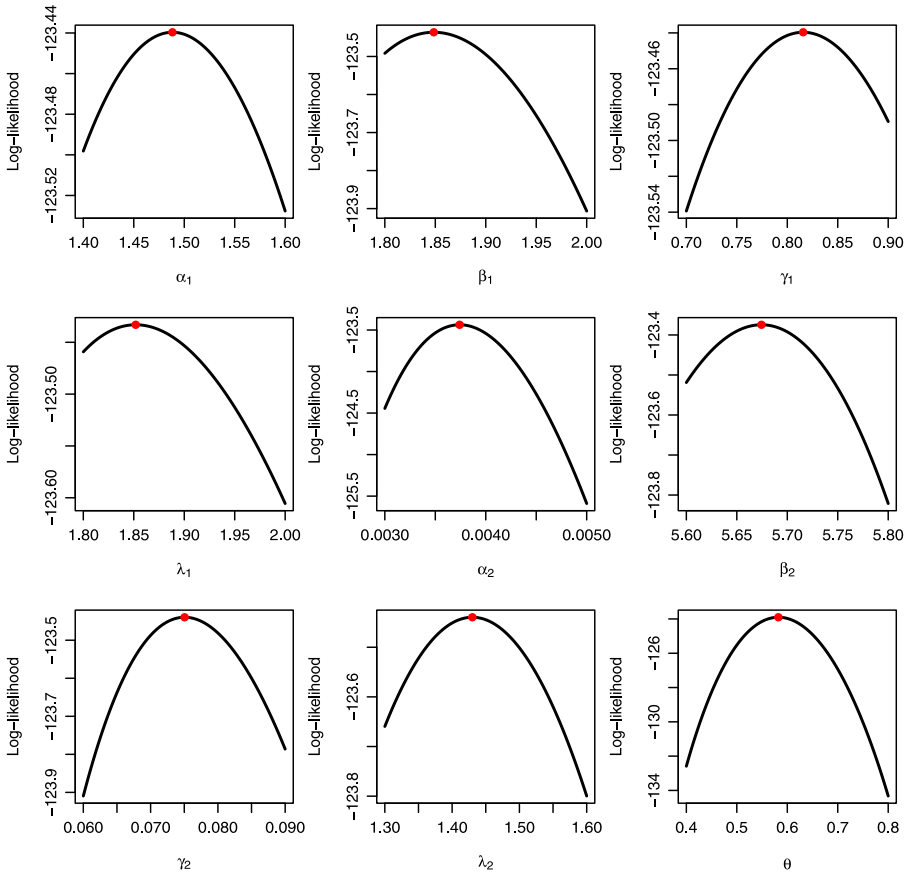


Fig. 9. The curves log-likelihood function of $\bigtriangledown = (a_1, a_2, b_1, b_2, c_1, c_2, d_1, d_2, \theta)$ for air quality data.

7. Conclusion

The present work in this article introduces the bivariate exponentiated additive Weibull distribution function whose marginals are exponentiated additive Weibull distributions. Since the joint distribution function and the joint probability density function are in closed forms, this distribution leads to a more flexible distribution capable of modeling various types of bivariate data. The proposed bivariate distribution's characteristics like conditional densities and product moments are explored. We generalized the results for the multivariate case. The ML estimators of the unknown parameters cannot be obtained in closed form, so we explore the numerical technique to get the ML estimates which works quite well. Additionally, the 95 percent confidence interval for the maximum likelihood estimates was obtained using the parametric bootstrap approach. The three real-life datasets from the failure times and warranty servicing times for nuclear power plants data, American football league, and air quality data were taken into consideration for comparison with the BGED, BGRD, BGaD, CBRD, and BVDD models to demonstrate the applicability of the proposed approach. The proposed bivariate exponentiated additive Weibull distribution outperformed the BGED, BGRD, BGaD, CBRD, and BVDD models for these three datasets in terms of log-likelihood, and AIC values.

CRediT authorship contribution statement

Neeraj Poonia: Conceptualization, Methodology, Software, Validation, Writing – original draft. **Deepak Prajapati:** Conceptualization, Methodology, Formal analysis, Investigation, Supervision, Writing – review & editing. **Sarita Azad:** Supervision, Project administration, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix

A.1. Proof of Theorem 2.3

Using the definition of mathematical expectation and Eq. (2.7), we get

$$E(x^l y^l) = \int_0^\infty \int_0^\infty x^l y^l \delta'(x)\delta'(y) \sum_{j=1}^{\infty} \binom{\theta}{j} (-1)^{j+1} j^2 e^{-j\phi(x,y)} dxdy$$
$$= \sum_{j=1}^{\infty} \binom{\theta}{j} (-1)^{j+1} j^2 I_{1j} I_{2j},$$

where $I_{1j} = \int_0^\infty x^l \delta'(x) e^{-j\delta(x)} dx$ and $I_{2j} = \int_0^\infty y^l \delta'(y) e^{-j\delta(y)} dy$. Now, using the Taylor series expansion of the function $e^{-jc_1 x^{d_1}} = \sum_{p=0}^{\infty} \frac{(-jc_1)^p}{p!} x^{pd_1}$. And put in I_{1j} we obtain

$$I_{1j} = \sum_{p=0}^{\infty} \frac{(-jc_1)^p}{p!} \int_0^\infty x^{l+pd_1} \delta'(x) e^{-ja_1 x^{b_1}} dx$$
$$= \sum_{p=0}^{\infty} \frac{(-jc_1)^p}{p!} \left[a_1 b_1 \int_0^\infty x^{l+pd_1+b_1-1} e^{-ja_1 x^{b_1}} dx \right.$$
$$\left. + c_1 d_1 \int_0^\infty x^{l+pd_1+d_1-1} e^{-ja_1 x^{b_1}} dx \right].$$

Using definition of Gamma function, we get

$$I_{1j} = \sum_{p=0}^{\infty} \frac{(-jc_1)^p}{p!} \left[\frac{a_1}{(ja_1)^{(l+pd_1+b_1)/b_1}} \Gamma\left(\frac{l+pd_1+b_1}{b_1}\right) \right.$$
$$\left. + \frac{c_1 d_1}{b_1 (ja_1)^{(l+pd_1+d_1)/b_1}} \Gamma\left(\frac{l+pd_1+d_1}{b_1}\right) \right].$$

Similarly,

$$I_{2j} = \sum_{q=0}^{\infty} \frac{(-jc_2)^q}{q!} \left[\frac{a_2}{(ja_2)^{(l+qd_2+b_2)/b_2}} \Gamma\left(\frac{l+qd_2+b_2}{b_2}\right) \right.$$
$$\left. + \frac{c_2 d_2}{b_2 (ja_2)^{(l+qd_2+d_2)/b_2}} \Gamma\left(\frac{l+qd_2+d_2}{b_2}\right) \right].$$

Hence above equations completes the proof.

A.2. Normal equations

$$\frac{\partial \mathscr{L}}{\partial \theta} = \frac{n}{\theta} - \sum_{j=1}^{n} \frac{e^{-\phi(x_j,y_j)}}{\left(1-\theta e^{-\phi(x_j,y_j)}\right)} + \sum_{j=1}^{n} \log\left(1-e^{-\phi(x_j,y_j)}\right) = 0. \tag{A.1}$$

$$\frac{\partial \mathscr{L}}{\partial a_1} = \sum_{j=1}^{n} \frac{b_1 x_j^{b_1-1}}{\delta'(x_j)} - \sum_{j=1}^{n} x_j^{b_1} + \sum_{j=1}^{n} \frac{x_j^{b_1} \theta e^{-\phi(x_j,y_j)}}{\left(1-\theta e^{-\phi(x_j,y_j)}\right)} + (\theta-2)$$
$$\times \sum_{j=1}^{n} \frac{x_j^{b_1} e^{-\phi(x_j,y_j)}}{\left(1-e^{-\phi(x_j,y_j)}\right)} = 0. \tag{A.2}$$

$$\frac{\partial \mathscr{L}}{\partial a_2} = \sum_{j=1}^{n} \frac{b_2 y_j^{b_2-1}}{\delta'(y_j)} - \sum_{j=1}^{n} y_j^{b_2} + \sum_{j=1}^{n} \frac{y_j^{b_2} \theta e^{-\phi(x_j,y_j)}}{\left(1-\theta e^{-\phi(x_j,y_j)}\right)} + (\theta-2)$$
$$\times \sum_{j=1}^{n} \frac{y_j^{b_2} e^{-\phi(x_j,y_j)}}{\left(1-e^{-\phi(x_j,y_j)}\right)} = 0. \tag{A.3}$$

$$\frac{\partial \mathscr{L}}{\partial c_1} = \sum_{j=1}^{n} \frac{d_1 x_j^{d_1-1}}{\delta'(x_j)} - \sum_{j=1}^{n} x_j^{d_1} + \sum_{j=1}^{n} \frac{x_j^{d_1} \theta e^{-\phi(x_j,y_j)}}{\left(1-\theta e^{-\phi(x_j,y_j)}\right)} + (\theta-2)$$
$$\times \sum_{j=1}^{n} \frac{x_j^{d_1} e^{-\phi(x_j,y_j)}}{\left(1-e^{-\phi(x_j,y_j)}\right)} = 0. \tag{A.4}$$

$$\frac{\partial \mathscr{L}}{\partial c_2} = \sum_{j=1}^{n} \frac{d_2 y_j^{d_2-1}}{\delta'(y_j)} - \sum_{j=1}^{n} y_j^{d_2} + \sum_{j=1}^{n} \frac{y_j^{d_2} \theta e^{-\phi(x_j,y_j)}}{\left(1-\theta e^{-\phi(x_j,y_j)}\right)} + (\theta-2)$$
$$\times \sum_{j=1}^{n} \frac{y_j^{d_2} e^{-\phi(x_j,y_j)}}{\left(1-e^{-\phi(x_j,y_j)}\right)} = 0. \tag{A.5}$$

$$\frac{\partial \mathscr{L}}{\partial b_1} = \sum_{j=1}^{n} \frac{a_1 x_j^{b_1-1} + a_1 b_1 x_j^{b_1-1} \log(x_j)}{\delta'(x_j)} + \sum_{j=1}^{n} \frac{\theta a_1 x_j^{b_1} \log(x_j) e^{-\phi(x_j,y_j)}}{\left(1-\theta e^{-\phi(x_j,y_j)}\right)}$$
$$- \sum_{j=1}^{n} a_1 x_j^{b_1} \log(x_j) + (\theta-2) \sum_{j=1}^{n} \frac{a_1 x_j^{b_1} \log(x_j) e^{-\phi(x_j,y_j)}}{\left(1-e^{-\phi(x_j,y_j)}\right)} = 0. \tag{A.6}$$

$$\frac{\partial \mathscr{L}}{\partial b_2} = \sum_{j=1}^{n} \frac{a_2 y_j^{b_2-1} + a_2 b_2 y_j^{b_2-1} \log(y_j)}{\delta'(y_j)} + \sum_{j=1}^{n} \frac{\theta a_2 y_j^{b_2} \log(y_j) e^{-\phi(x_j,y_j)}}{\left(1-\theta e^{-\phi(x_j,y_j)}\right)}$$
$$- \sum_{j=1}^{n} a_2 y_j^{b_2} \log(y_j) + (\theta-2) \sum_{j=1}^{n} \frac{a_2 y_j^{b_2} \log(y_j) e^{-\phi(x_j,y_j)}}{\left(1-e^{-\phi(x_j,y_j)}\right)} = 0. \tag{A.7}$$

$$\frac{\partial \mathscr{L}}{\partial d_1} = \sum_{j=1}^{n} \frac{c_1 x_j^{d_1-1} + c_1 d_1 x_j^{d_1-1} \log(x_j)}{\delta'(x_j)} + \sum_{j=1}^{n} \frac{\theta c_1 x_j^{d_1} \log(x_j) e^{-\phi(x_j,y_j)}}{\left(1-\theta e^{-\phi(x_j,y_j)}\right)}$$
$$- \sum_{j=1}^{n} c_1 x_j^{d_1} \log(x_j) + (\theta-2) \sum_{j=1}^{n} \frac{c_1 x_j^{d_1} \log(x_j) e^{-\phi(x_j,y_j)}}{\left(1-e^{-\phi(x_j,y_j)}\right)} = 0. \tag{A.8}$$

$$\frac{\partial \mathscr{L}}{\partial d_2} = \sum_{j=1}^{n} \frac{c_2 y_j^{d_2-1} + c_2 d_2 y_j^{d_2-1} \log(y_j)}{\delta'(y_j)} + \sum_{j=1}^{n} \frac{\theta c_2 y_j^{d_2} \log(y_j) e^{-\phi(x_j,y_j)}}{\left(1-\theta e^{-\phi(x_j,y_j)}\right)}$$
$$- \sum_{j=1}^{n} c_2 y_j^{d_2} \log(y_j) + (\theta-2) \sum_{j=1}^{n} \frac{c_2 y_j^{d_2} \log(y_j) e^{-\phi(x_j,y_j)}}{\left(1-e^{-\phi(x_j,y_j)}\right)} = 0. \tag{A.9}$$

A.3. MLE code for given data

Let data point (x, y) is coming from Failure and Warranty servicing time data/ American Football league/Air Quality data, where

x = failure times in days/ time for the first points earned by kicking/ mean ozone in parts per billion.

y = warranty servicing times in days/ time for moving the ball into the end zone/solar radiation.

Then the R code to obtain the MLE for these datasets is given below:

```
n = length(x)
nll.beaw = function(pv){
 al1 = pv[1]
```

```
 be1 = pv[2]
 ga1 = pv[3]
 la1 = pv[4]
 al2 = pv[5]
 be2 = pv[6]
 ga2 = pv[7]
 la2 = pv[8]
 th = pv[9]

 ll = n*log(th) + sum( log(al1*be1*(x^(be1-1)) +
 ga1*la1*(x^(la1-1))) ) + sum( log(al2*be2*
 (y^(be2-1)) + ga2*la2*(y^(la2-1))) ) - sum(al1*x^be1
 + ga1*x^la1 + al2*y^be2 + ga2*y^la2) + sum( log(1 - th*
 exp( -(al1*x^be1 +  ga1*x^la1 + al2*y^be2 +
 ga2*y^la2) )) )+ (th - 2)*sum( log(1 - exp(-(al1*x^be1
 +  ga1*x^la1 + al2*y^be2 + ga2*y^la2))) )
 return(-ll)
}

pv = c(0.1,0.1,0.1,0.1,0.1,0.1,0.1,0.1,0.1)
optim(fn = nll.beaw, par = pv,method= "L-BFGS-B",
lower = c(0.000001,0.000001,
0.000001,0.000001,0.000001,0.000001,0.000001,
0.000001,0.000001),upper = c(100,100,
100,100,100,100,100,100,100,1))
```

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